Writing Equations from Word Situations
Answer Key
Grade 7

Quick Answers

1. 18
2. 90
3. 74
4. (b) kilometres travelled = 8, —
5. 20
6. 9
7. 13
8. (b) plants in one row = 9, —
9. 160
10. (b) adult tickets sold = 80, —

What is verified

7 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 4, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 7
6.EE.B.7 – *problems 1, 2, 3, 5, 7*
7.EE.B.4 – *problems 4, 6, 8, 9, 10*
Difficulty 1–5 of 5 across 14 tagged checks.

1. Maya had s stickers, was given 14 more, and had 32.
Solution: "Given 14 more" adds to the start, so $s + 14 = 32$. Subtract 14 from both sides: $s = 32 - 14 = 18$ stickers. $\boxed{s = 18}$

2. A bag of m marbles shared equally by six friends gives 15 each.
Solution: Sharing equally divides, so $\frac{m}{6} = 15$. Multiply both sides by 6: $m = 15 \times 6 = 90$ marbles. $\boxed{m = 90}$

3. A tank of w litres lost 26 litres and 48 were left.
Solution: Draining subtracts, so $w - 26 = 48$. Add 26 to both sides: $w = 48 + 26 = 74$ litres. $\boxed{w = 74}$

4. Taxi: 3 dollars to start plus 2 dollars per kilometre; the trip cost 19 dollars. Write and solve for k.

Solution: (a) **Instructor-judged.** The distance charge is 2 dollars for each of k kilometres, which is $2k$, and the 3 dollar start charge is paid once: $2k + 3 = 19$. Accept any equivalent arrangement — $3 + 2k = 19$, $19 = 2k + 3$, $19 - 3 = 2k$. An expression with no equals sign, or an equation that multiplies the start charge by the distance, is not correct.

(b) Subtract the start charge, then divide by the rate: $2k = 16$, so $k = 8$ kilometres.

$\boxed{k = 8}$

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5. A rope of r cm cut into 8 equal pieces of 2.5 cm each.

Solution: Cutting into 8 equal pieces divides, so $\frac{r}{8} = 2.5$. Multiply both sides by 8: $r = 2.5 \times 8 = 20$ centimetres. $\boxed{r = 20}$

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6. Swim club: 12 dollars to join plus 5 dollars a month, 57 dollars paid.

Solution: The joining fee is paid once and the 5 dollars repeats each month, so $5n+12 = 57$. Subtract 12: $5n = 45$. Divide by 5: $n = 9$ months. $\boxed{n = 9}$

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7. Temperature t fell 17 degrees to reach -4 degrees.

Solution: Falling subtracts, so $t - 17 = -4$. Add 17 to both sides: $t = -4 + 17 = 13$ degrees Celsius. The answer is above zero even though the finish is below it, because the fall was larger than the finishing temperature. $\boxed{t = 13}$

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8. Six equal rows of p plants plus 4 extra plants make 58.

Solution: (a) **Instructor-judged.** Six rows of p plants is $6p$, and the 4 potted plants are added once: $6p + 4 = 58$. Accept $4 + 6p = 58$ or $58 - 4 = 6p$. Writing $6(p + 4) = 58$ is not correct — that puts four extra plants in every row.

(b) Subtract 4: $6p = 54$. Divide by 6: $p = 9$ plants in each row. $\boxed{p = 9}$

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9. Flour is 3 times the sugar; together they weigh 640 grams. Solve for the sugar g.

Solution: The sugar is g grams and the flour is $3g$ grams, and the two together are 640 grams: $g + 3g = 640$. Collect like terms: $4g = 640$. Divide by 4: $g = 160$ grams of sugar (and 480 grams of flour). $\boxed{g = 160}$

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10. Adult tickets at 8 dollars, 60 child tickets at 5 dollars, 940 dollars in all. Solve for the number a of adult tickets.

Solution: (a) **Instructor-judged.** The adult money is $8a$ and the child money is fixed at $5 \times 60 = 300$ dollars, so $8a + 300 = 940$. Accept the unsimplified form $8a + 5(60) = 940$.

Giving the child tickets their own letter is not correct here: that number is already known.

(b) Subtract 300: $8a = 640$. Divide by 8: $a = 80$ adult tickets. $\boxed{a = 80}$

(c) **Instructor-judged.** A correct response gives a meaning to each piece: a is the number of adult tickets, the 8 multiplying it is the price of one adult ticket in dollars, the 300 (or 5×60) is the money already taken from the child tickets, and 940 is the whole amount collected. Full credit needs a meaning for the letter, for the multiplied term, and for the constant term.
